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1. INTRODUCTION

Agriculture continues to be one of the most important sectors in India, yet the earning potential of farmers remains significantly lower compared to the effort they invest. A major contributor to this problem is the traditional marketing chain that involves multiple middlemen. These intermediaries often take a large share of profit, leaving farmers with minimal returns while consumers still pay high prices. With the increasing availability of digital platforms and improved internet access, it has become possible to reduce these inefficiencies by enabling direct communication between farmers and consumers.

1.1 Problem Definition

In many rural regions, farmers rely heavily on traditional market systems where they sell their produce to middlemen who later supply it to wholesale markets or retailers. While this system has existed for years, it is highly inefficient. Farmers often lack bargaining power and have limited knowledge of market trends, leading to unfair pricing. Consumers, on the other hand, buy the same products at much higher prices due to added commissions.

1.2 Objectives

The main objective of AgroTrade is to create a digital, transparent, and fair marketplace. Specifically, the system aims to: Provide an online space where farmers can list their produce easily and update availability and pricing whenever needed, Allow consumers to browse fresh agricultural products without relying on offline markets, Encourage direct communication between farmers and customers by avoiding agents or brokers and Build a platform that is simple enough for new users to understand and use without training.

1.3 Scope of the Project

The AgroTrade system covers the essential features required for small-scale digital trading. It supports three main user roles: farmer, consumer, and admin. Farmers can manage products, consumers can buy them, the structure can be expanded later for statewide or national use.

Features like delivery management, automatic price suggestions, or product-quality checks can also be added in future versions.

1.4 Advantages

The platform offers several benefits: Farmers control the price of their produce based on effort and market demand, Buyers can see exact product details, availability, and pricing and Simple navigation allows even first-time users to operate the system comfortably.

1.5 Limitations

While AgroTrade is helpful, certain limitations still exist:

- The system depends entirely on internet connectivity.
- A delivery or logistics system is not integrated, so buyers and farmers must coordinate themselves.

2. LITERATURE SURVEY

Digital agriculture solutions have been studied extensively in recent years, especially in relation to improving farmer incomes and market efficiency. Many research articles and government reports highlight that traditional agricultural markets involve multiple layers of intermediaries, reducing profitability. Studies also show that farmers struggle with market access, price prediction, and storage management, which further affects earnings.

2.1 Existing Systems

Several existing systems such as eNAM, AgriBazaar, and AgroWave already provide online agriculture trading. These platforms usually offer: Price information Product listing, Online bidding and Bulk trading options

Although these systems are effective, they primarily target large-scale markets or require farmers to follow specific registration processes. Many farmers find it difficult to upload documents, verify identity, or navigate complicated dashboards. Additionally, some services still involve agents or commission-based transactions.

2.2 Comparison of Existing Systems and AgroTrade

AgroTrade differs from existing platforms mainly in its simplicity. It does not require complicated verification steps or payment gateways. Farmers only need basic details to create an account and list their products. The focus is on making a beginner-friendly marketplace rather than a highly commercial one.

2.3 Literature Findings

Studies indicate that digital marketplaces significantly improve farmers' incomes by allowing them to reach wider audiences. Findings also support that user-friendly digital tools increase farmer participation compared to complex commercial portals.

2.4 Comparative Analysis

A short analysis shows that: Existing systems target large-scale markets, while AgroTrade is designed for simple local use, Many platforms include service charges, while AgroTrade is completely free. And AgroTrade focuses on direct communication and transparency.

3.REQUIREMENTS SPECIFICATION

A successful system needs clear requirements. AgroTrade's requirements include what the system should do (functional requirements) and how it should behave (non-functional requirements). The requirements also include details related to performance, database structure, and user restrictions.

3.1 Functional Requirements

Farmers must be able to add, edit, and delete products. Each product will include details such as crop name, price per unit, description, and availability. This ensures farmers can control their listings and update their stock regularly. The system must allow farmers, consumers, and admins to register and log in securely. Passwords should be stored safely in the database. Each user should have access only to their respective dashboard and functionalities.

3.2 Non-Functional Requirements

Farmers must be able to add, edit, and delete products. Each product will include details such as crop name, price per unit, description, and availability. No user should be able to access someone else's dashboard. Forms must validate input to prevent SQL injection or data corruption. The UI must be readable and simple. Buttons and text sizes should be clear, especially because farmers may not be familiar with complex systems.

3.3 Database Requirements

The database should store user details, product information, login data, and category details. The tables must have proper primary and foreign keys to maintain data integrity.

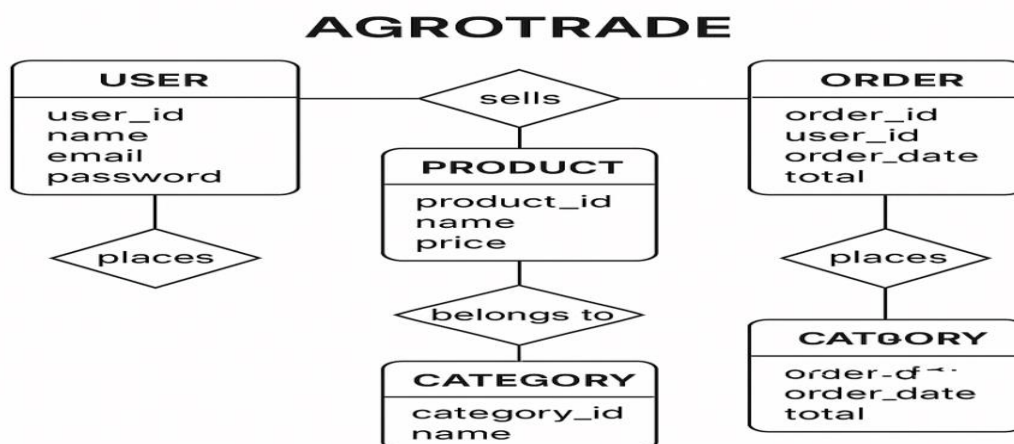


Fig 3.1 E-R Diagram

3.4 Software Requirements

XAMPP for running MySQL and Apache, PHP as backend scripting language, HTML/CSS for UI and JavaScript for client-side validation

4. TECHNOLOGY STACK, IMPLEMENTATION & DESIGN

In developing AgroTrade, I chose a technology stack that is simple, lightweight, and easy to maintain, making it suitable for academic projects as well as small-scale real-world applications. The aim was to create a platform where the technology does not complicate the user experience but instead supports smooth navigation, secure data handling, and reliable system behavior. Each tool used in this project has been selected specifically because it fits the needs compatibility with MySQL-based backend systems. of how AgroTrade was built from start to finish.

4.1 Technology Stack Overview

HTML forms the backbone of the user interface. It structures all the pages, including registration forms, product lists, and admin panels. Every input on the platform is created using HTML. Its simplicity ensures that the website remains accessible and lightweight, especially for users with slower internet connections.

CSS is used to give the platform a clean and attractive look. By adding colors, spacing, borders, and fonts, CSS improves the visual appeal of the pages and enhances overall usability. In AgroTrade, especially for users on mobile devices or those who are not familiar with complex web layouts.

JavaScript adds interactivity to the system. It is mainly used for front-end validation of forms, and responsive behavior. JavaScript ensures that incorrect inputs are identified before sending data to the server, improving both usability and security.

MySQL is used to store all essential data, such as user accounts, product listings, categories, and admin information. Tables are created with proper relationships, to maintain data integrity. SQL queries and delete data according to user actions.

4.2 System Implementation and Architecture

The implementation of AgroTrade happened in multiple phases, each focusing on structuring the system so that every module worked independently yet remained connected to the database and backend. The architecture follows a three-layer model:

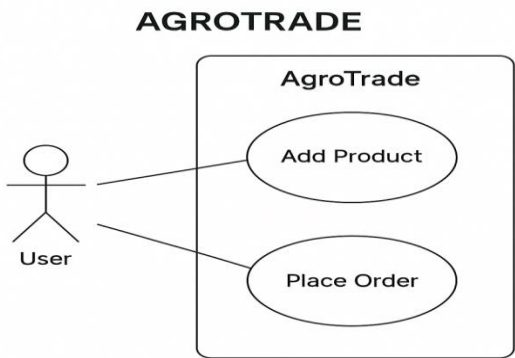


Fig. 4.1 Use-case diagram

Each page uses a simple layout: Header (navigation bar), Body (main content) and Footer (copyright) .The layout is consistent across the platform for familiarity.



Fig. 4.2 Sequence diagram

MySQL tables were designed to support modularity. Important tables include: **users** (stores user login info), **farmers** (farmer profile details), **products** (crop details, prices, images), **categories** (classification of crops), **admin** (system administrators). Primary keys and foreign keys maintain relationships between tables.

5. DESIGN AND USER INTERFACE(UI)

The design and implementation of the AgroTrade platform focus on converting planned system features into a functional, user-friendly application. A simple, responsive, and clean UI was prioritized so that farmers, buyers, and admins can use the platform without technical difficulty. The following sections describe the tools used, the implementation process, role-based access structure, and key UI elements of AgroTrade.

5.1 Implementation Steps

The system was implemented by setting up XAMPP, creating the database, and building the frontend pages with validations. PHP handled backend functions like login and product management, and after integrating all modules, the platform was fully tested before deployment.

5.3 Webpage Overview

The AgroTrade website consists of several key pages that collectively support smooth agricultural to a simple and efficient workflow. This structure ensures that users can find information easily, complete actions, and interact with the platform in a straightforward manner.

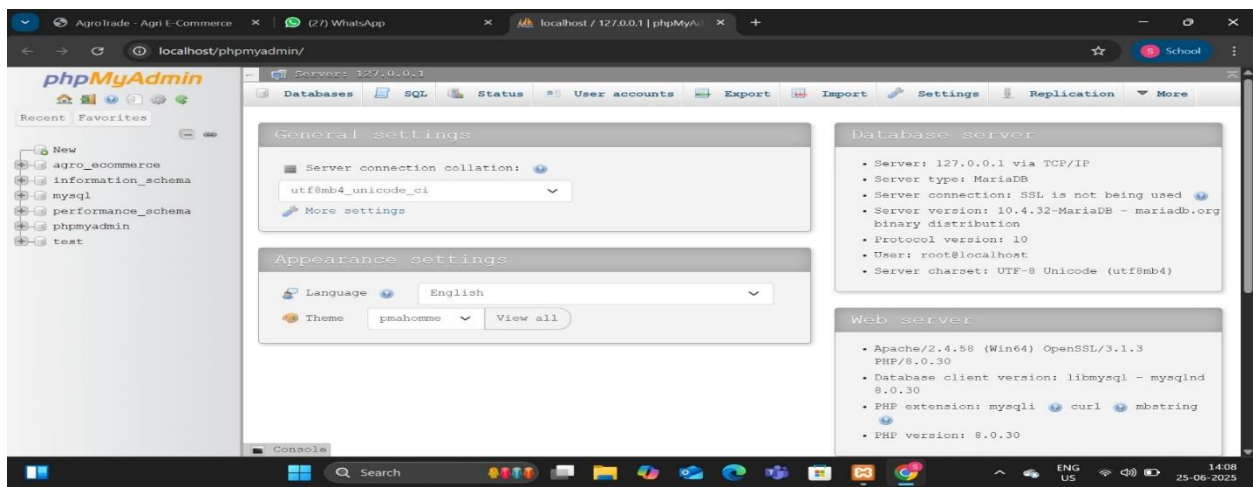
5.3.2 Home page

In Fig. 5.1, the homepage of the AgroTrade platform displays a simple navigation bar with options like Home, Products, Sell, Mandi Rates, Login, and Cart. The layout is clean, intuitive, and designed to support smooth agricultural trading for all users.



Fig. 5.1 Home page

5.3.2 Database page



5.2 Database page

Fig. 5.2 shows the phpMyAdmin interface used to manage the MySQL/MariaDB database for the AgroTrade project. The database *agro_ecommerce* contains tables for users, products, orders, and transactions. This setup is powered by XAMPP, allowing the developer to run and test the system locally before deployment.

CONCLUSION

The AgroTrade website successfully provides a digital solution to bridge the gap between farmers, buyers, and traders by offering a transparent, efficient, and user-friendly agricultural marketplace. By leveraging modern web technologies and a structured system design, the platform enables users to list, browse, and trade agricultural products with ease. Features like role-based access, real-time product listings, secure transactions, and feedback systems ensure a smooth and trustworthy user experience. The integration of tools like HTML, CSS, JavaScript, PHP, MySQL, and AJAX has helped in building a responsive and interactive interface suitable for all stakeholders.

Overall, AgroTrade not only simplifies agricultural trading but also empowers rural communities by improving market access and reducing dependency on intermediaries. The platform is designed to be scalable and maintainable, allowing future enhancements such as mobile app integration, AI-based price predictions, and supply chain tracking. In conclusion, AgroTrade stands as a step forward in digital agriculture, promoting fair trade, reducing wastage, and contributing to the sustainable development of the farming ecosystem.